GOVERNMENT OF INDIA

Ministry of Science & Technology

Innovating for a Better Tomorrow

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SHRI NARENDRA MODI

HONORABLE PRIME MINISTER, INDIA



MESSAGE

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HONORABLE MINISTER,

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MESSAGE

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DIRECTOR GENERAL, CHAMBER OF COMMERCE AND INDUSTRIES OF INDIA

MESSAGE

Science, technology, and innovation have become the driving forces behind economic competitiveness, industrial transformation, and sustainable national development. The Ministry of Science and Technology plays a pivotal role in advancing scientific research, fostering technological innovation, promoting entrepreneurship, strengthening research institutions, and supporting the commercialization of emerging technologies that contribute to India's global leadership.

The Chamber of Commerce and Industries of India (CCI India) recognizes that innovation-led growth requires stronger collaboration between government, industry, academia, startups, investors, and research institutions. Through policy advocacy, innovation platforms, research partnerships, technology commercialization, startup incubation, investment facilitation, and capacity-building initiatives, CCI India remains committed to supporting the Ministry's vision of building a globally competitive innovation ecosystem.

This Knowledge Paper provides comprehensive insights into the Ministry's strategic priorities, research ecosystem, flagship initiatives, policy landscape, and emerging opportunities across biotechnology, advanced materials, artificial intelligence, quantum technologies, space applications, deep-tech innovation, climate technologies, and scientific entrepreneurship. It also identifies strategic areas where collaborative partnerships can accelerate research commercialization and industrial innovation.

The publication is intended to serve as a strategic reference for policymakers, scientists, researchers, entrepreneurs, technology companies, investors, academic institutions, innovation hubs, and development organizations seeking to contribute to India's scientific advancement.

As India moves towards becoming a global innovation leader, science and technology will continue to shape the nation's future. CCI India remains committed to partnering with the Ministry of Science and Technology in advancing research excellence, technological innovation, and knowledge-driven economic development.

Munindra Kumar
Director General
CCI India

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Preface

Science, technology and innovation have become central to how nations compete and achieve sustainable growth. Within the Government of India's institutional framework, the Ministry of Science and Technology holds a uniquely strategic position, serving as a funder of fundamental research, a promoter of applied and industrial technologies, and an increasingly significant catalyst for deep-tech and biotechnology innovation through initiatives such as the Anusandhan National Research Foundation and the Research, Development and Innovation Fund. Few ministries integrate scientific research, industrial innovation and technology financing under a single administrative umbrella to this extent.

India's innovation ecosystem has made notable progress, rising from 81st to 38th in the Global Innovation Index over the past decade and emerging as the world's third-largest producer of scientific publications. However, research and development expenditure remains around 0.64 per cent of GDP, significantly below the 2–5 per cent invested by leading innovation economies. Achieving the ambitions of Viksit Bharat 2047 will therefore require stronger collaboration between government, industry, academia and investors to mobilise resources, expertise and innovation at scale.

This knowledge paper, prepared by CCI India, presents an evidence-based case for a structured institutional partnership between the Ministry of Science and Technology and CCI India. Drawing on the Ministry's budget documents, scheme guidelines, parliamentary disclosures and performance data, it examines current priorities, identifies implementation challenges, and highlights areas where CCI India can contribute through industry mobilisation, knowledge partnerships and collaborative programme delivery without duplicating the Ministry's statutory responsibilities.

CCI India's objective is to move beyond periodic engagement towards a long-term partnership model built on shared priorities, joint governance mechanisms and measurable outcomes. The paper is intended to provide senior policymakers and stakeholders with a concise, evidence-driven rationale for why this collaboration is both timely and strategically important.

The intended audience includes the Union Minister and Minister of State for Science and Technology, senior officials of the Department of Science and Technology, Department of Biotechnology and Department of Scientific and Industrial Research, members of NITI Aayog, policymakers across allied ministries, industry leaders, investors, multilateral development institutions and research organisations engaged in advancing India's innovation ecosystem.

Chamber of Commerce and Industry (CCI India)

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About CCI India

CCI India (The Chamber of Commerce and Industry of India) is a premier national business chamber dedicated to fostering economic growth, industry development, entrepreneurship, innovation, and investment promotion across India. As the voice of Indian business and industry, CCI India serves as a bridge between government, industry, startups, academia, investors, and civil society, facilitating meaningful collaboration for nation-building and sustainable development.

Through policy advocacy, business facilitation, research, stakeholder engagement, capacity building, and strategic partnerships, CCI India works to create an enabling environment for enterprises of all sizes. The Chamber actively supports key national initiatives such as Startup India, Make in India, Digital India, Atmanirbhar Bharat, Skill India, and Viksit Bharat 2047 by connecting innovative enterprises with government opportunities and promoting inclusive economic development.

CCI India operates through sector-specific councils, industry forums, knowledge platforms, and state-level engagements that address emerging opportunities and challenges across sectors including manufacturing, agriculture, technology, healthcare, education, infrastructure, sustainability, MSMEs, and startups. The Chamber regularly organizes policy dialogues, investment summits, business delegations, workshops, conferences, and knowledge-sharing initiatives to strengthen India's economic ecosystem.

With a strong commitment to innovation, entrepreneurship, and public-private collaboration, CCI India strives to empower enterprises, facilitate investment, enhance competitiveness, and contribute to India's journey toward becoming a globally competitive and economically resilient nation. Its role as a knowledge partner and industry catalyst enables ministries, departments, state governments, startups, and businesses to collaborate effectively in achieving national development objectives.

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Section 1: Introduction to the Ministry

The Ministry of Science and Technology was constituted in May 1971, when the Department of Science and Technology was carved out as a dedicated nodal department, reflecting the political weight assigned to scientific capacity-building in the early decades of post-Independence planning. Over the subsequent five decades, the Ministry's footprint expanded through the establishment of the Department of Biotechnology in 1986 and the formal notification of the Department of Scientific and Industrial Research in 1985, giving the Ministry its present three-department structure. This evolution mirrors a broader pattern in Indian governance: as scientific disciplines matured into distinct policy domains requiring specialised regulatory and funding architecture, the Ministry chose to internally federate rather than fragment, preserving a single ministerial point of accountability while allowing each department operational autonomy suited to its domain.

The Ministry's governance role is best understood as that of a system-architect rather than a service-delivery department in the conventional sense. It does not, for the most part, deliver services directly to citizens; instead, it designs funding instruments, sets research priorities, accredits and regulates scientific institutions, and creates the enabling conditions under which universities, national laboratories, start-ups and private industry conduct research and translate it into deployable technology. This positions the Ministry as an upstream actor whose effectiveness is measured less by direct beneficiary counts and more by downstream indicators: publication output, patent filings, technology transfer volumes, start-up formation in deep-tech sectors, and the eventual commercial and social application of publicly funded research.

Constitutionally, scientific and technical education and research fall predominantly within the Union's domain under the Seventh Schedule's Union List, granting Parliament and the Union Executive primary legislative and executive competence over institutions of national importance in science and technology. This constitutional placement is significant for any partnership model: it means that the Ministry's authority to enter into structured collaborations, sign memoranda of understanding, and direct funding through statutory bodies such as the Anusandhan National Research Foundation is exercised with a degree of centralised coherence that is less commonly available in subjects where coordination across multiple state governments would otherwise be required.

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National significance follows directly from this constitutional centrality. The Ministry's three departments collectively underwrite the basic research base that other mission-mode ministries depend upon, support the biotechnology and pharmaceutical value chains that anchor India's position as a global vaccine and generic-drug supplier, and provide the technology-transfer infrastructure through which laboratory discoveries in the Council of Scientific and Industrial Research network reach industrial application. As India pursues a target of becoming a developed economy by 2047, the Ministry's role shifts from being a quiet enabler of academic science to being a frontline instrument of economic strategy, particularly in deep-tech, quantum technology, semiconductor design, biomanufacturing and artificial intelligence-linked research domains where global competitiveness is increasingly determined by the speed and scale of public-private research collaboration.

Section 2: Historical Timeline of the Ministry

The Ministry's institutional evolution has proceeded through distinct waves, each triggered by a shift in national priority: nation-building science in the 1970s and 1980s, liberalisation-era industrial research in the 1990s, biotechnology and intellectual-property modernisation in the 2000s, and mission-mode deep-tech financing from 2020 onward. Tracking this sequence matters strategically because each wave left behind institutional residue, in the form of statutory bodies, funding instruments and laboratory networks, that still structures how the Ministry operates today.

Year	Milestone	Institutional Significance
1942	Council of Scientific & Industrial Research (CSIR) established	Created India's first national laboratory network, later placed under DSIR
1971	Department of Science & Technology (DST) established	Formed the Ministry's nodal coordinating department for national S&T policy
1985	Department of Scientific & Industrial Research (DSIR) notified	Gave CSIR's 38-plus national laboratories a dedicated administrative home
1986	Department of Biotechnology (DBT) established	Created a standalone department for life-sciences research and biomanufacturing
2008	Science and Engineering Research Board (SERB) Act passed	Introduced India's first statutory, peer-reviewed competitive research-grant body
2016	Start-up India and Atal Innovation Mission launched	Embedded entrepreneurship and incubation into the national innovation pipeline

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